Jiamao Liu

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EDUCATION

Xiamen University Xiamen, China
M.S., Bioinformatics, Average Score: 85/100 Sept. 2024 – Present

Coursework: Scientific Writing (89), Molecular and Cellular Biology (87), Frontiers in Life Sciences Research (96)

Xiamen University Xiamen, China
B.S., Biological Science, Average Score: 80/100 Sept. 2020 – Jun. 2024

Coursework: Biostatistics (97), Cell and Microscope Experiment (94), AI Application and Practice (92)

RESEARCH INTERESTS

Single-cell and spatial multi-omics; epigenomics; enhancers; deep learning for genomics

PUBLICATIONS

Co-first authors; ∗ Corresponding authors

Published

- Hong, D.#, Shu, M.#, **Liu, J.#**, Liu, L., Cheng, H., Zhu, M., Du, Y., Xu, B., Hu, D., Liu, Z., Zhao, Y., Dai, J.*, Lu, F.*, Huang, J.* (2025). Divergent combinations of enhancers encode spatial gene expression. ***Nature Communications***, 16, 5091.

In Preparation

- Zhu, M.#, Zha, Z.#, Xuan, J., Zhao, H., Dai, J., **Liu, J.**, Xia, Y., Lin, H., Yang, Y., Chen, F., Lin, S., Liu, X., Wu, L., Zhu, Z., Huang, J.*, Zhang, H.*, Yang, C.*. Super-CUT&Tag: a sensitive, spatially resolved approach for epigenomic profiling in tissue sections. *Nature Structural & Molecular Biology* (**in revision**).
- Shu, M.#,*, **Liu, J.#**, Wang, M.#, Chen, Z.#, Xu, C., Zhang, Q., Hong, D., Zhang, Y., Li, Y., Kuang, W., Lin, H., Wei, Y., Liu, L., Xu, B., Cheng, H., Duan, L., Xue, X., Chen, B., Xiao, Z., Zhao, Y., Huang, J.*, Dai, J.*, Lu, F.*. Decoding the spatiotemporal and epigenetic principles underlying human spinal cord development. *Nature Cell Biology* (**under review**).

RESEARCH EXPERIENCE

Spatial Enhancer Code Reveals Regional Gene Regulation Sept. 2023 – Jun. 2025
Co-first author, **Nature Communications** (2025)

- Co-developed eSpatial, a computational framework integrating spatial transcriptomic and chromatin accessibility data to infer spatially resolved enhancer-gene regulatory relationships and identify divergent enhancer combinations.
- Analyzed spatial ATAC-seq of the mouse brain by integrating single-cell ATAC-seq annotations and quantifying cell-type-specific enhancer accessibility along anatomical axes, revealing spatially divergent enhancer activity within individual neuronal subtypes.
- Contributed to data visualization, figure preparation, biological interpretation, manuscript writing, and revision.

Multi-omics Atlas of Human Spinal Cord Development Mar. 2024 – Jun. 2026
Co-first author

- Led computational analysis of single-nucleus multiome and spatial transcriptomic data across human spinal cord development, including cell-type annotation, multi-omic integration, developmental trajectory inference, and spatial mapping.

- Designed and performed enhancer-gene linkage, chromatin priming, transcription factor footprinting, and gene regulatory network analyses to characterize developmental regulatory dynamics.
- Generated the majority of computational results and figures; contributed substantially to manuscript writing and submission.

Spatial GWAS Enrichment Analysis for Tissue Epigenomics Jun. 2025 – Mar. 2026
Independent computational project

- Performed cell-type-specific GWAS enrichment analysis using spatial epigenomic data to map trait-associated regulatory signals across anatomically defined tissue regions.
- Conducted statistical analysis and visualization, prepared figures, and contributed to biological interpretation and manuscript writing.

Non-invasive Embryo Assessment through cfDNA Methylation Profiling Nov. 2025 – Present
Independent computational project

- Established a computational workflow for analyzing low-input cfDNA methylation data from spent embryo culture medium, including quality control, methylation profiling, differential analysis, and CNV analysis.
- Implemented and evaluated DECENT/NICE-based decontamination strategies to improve recovery of embryo-derived cfDNA signals.
- Currently identifying pregnancy outcome-associated methylation biomarkers and developing predictive models for non-invasive embryo assessment.

Synthetic-Supervised Neural Network for Spatial Enhancer-Gene Regulation Jun. 2026 – Present
Primary Researcher

- Developing a synthetic-supervised neural network to characterize spatial discrepancies between enhancer accessibility and gene expression patterns from spatial multi-omic data.
- Applying the model to identify complex spatial enhancer-gene regulatory patterns and prioritize genes exhibiting spatially heterogeneous regulation across large-scale datasets.
- Incorporating transcription factor (TF)-based interpretation to identify candidate TFs underlying distinct spatial enhancer–gene regulatory patterns.

AWARDS

- National Scholarship for Graduate Students, China (Two-time recipient; Top 2%; 2025, 2026)
- Outstanding Student Award, Xiamen University (Top 5%, 2025)
- Software Copyright: *eSpatial: Decoding Spatial Regulatory Mechanisms of Enhancers* (2025)

PRESENTATIONS

- **Poster**, 5th Annual Symposium of the Fujian Bioinformatics Society, First Prize, 2025.
- **Poster**, Xiamen Society for Immunology Annual Meeting, 2026.

SKILLS

Computing: Python, R, PyTorch, Linux, HPC, Git, Docker
Bioinformatics: Scanpy, Seurat, SnapATAC2, ArchR, Signac, SCENIC+, Bismark, deepTools
Languages: Chinese (Native), English (Professional Working Proficiency)